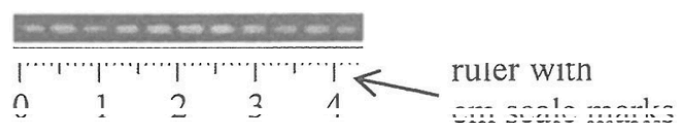


*6. (a) A laser beam is directed perpendicularly towards a double slit of separation $a = 0.3$ mm. The pattern of bright spots projected on a screen 1.8 m away from the slits is shown in Figure 6.1.

Figure 6.1



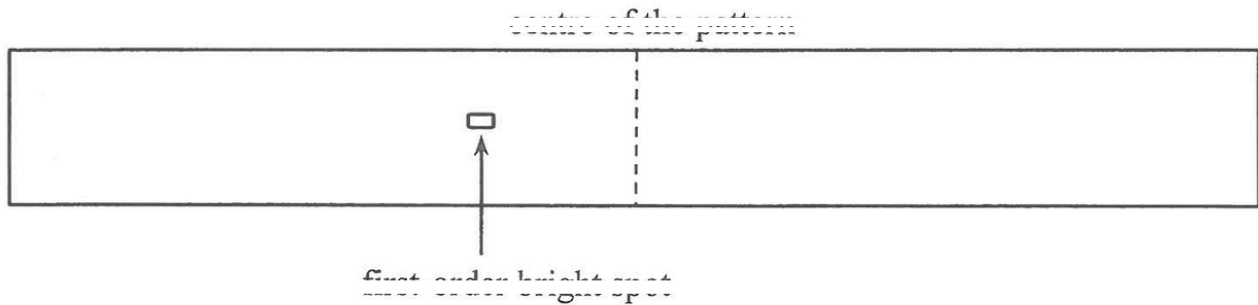
- (i) Find the wavelength of the laser beam. (3 marks)

- (ii) Explain why the slit width has to be very narrow in order for the above pattern to be observed. (2 marks)

Answers written in the margins will not be marked.

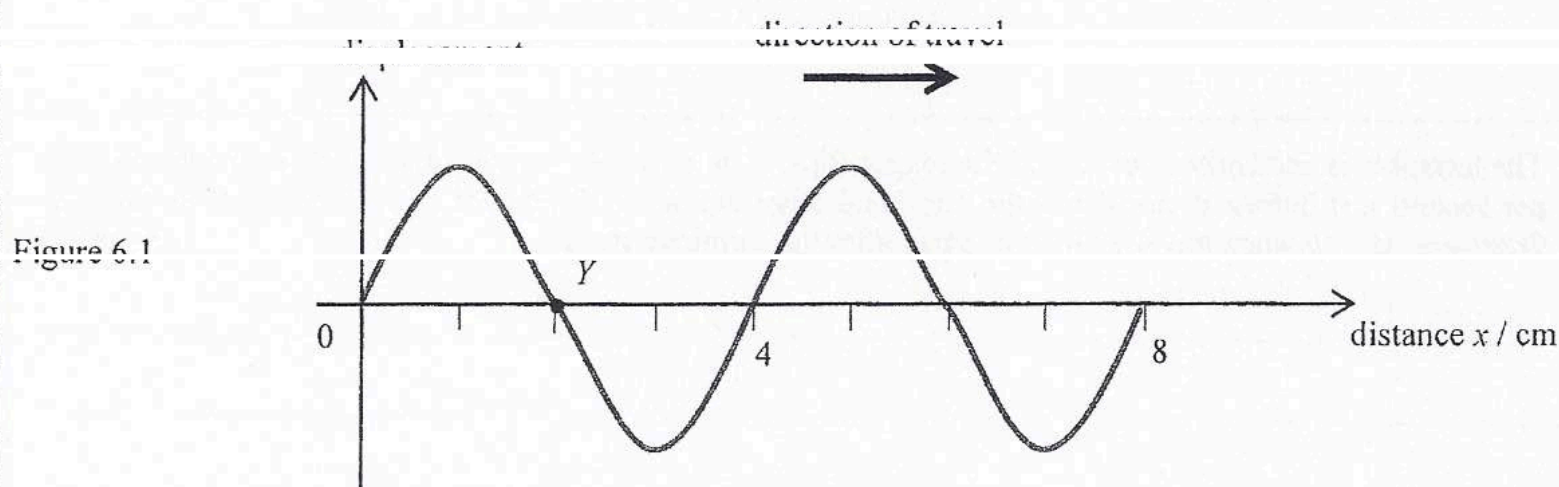
- (b) The double slit is now replaced by a diffraction grating of 500 lines per mm.
- (i) Find the separation between the central bright spot and first-order bright spot of the pattern on the screen for the same experimental settings. (3 marks)

- (ii) Sketch the pattern, up to the second-order, that you would expect to see on the screen when using this diffraction grating. A first-order bright spot has already been drawn for you. (2 marks)



Answers written in the margins will not be marked.

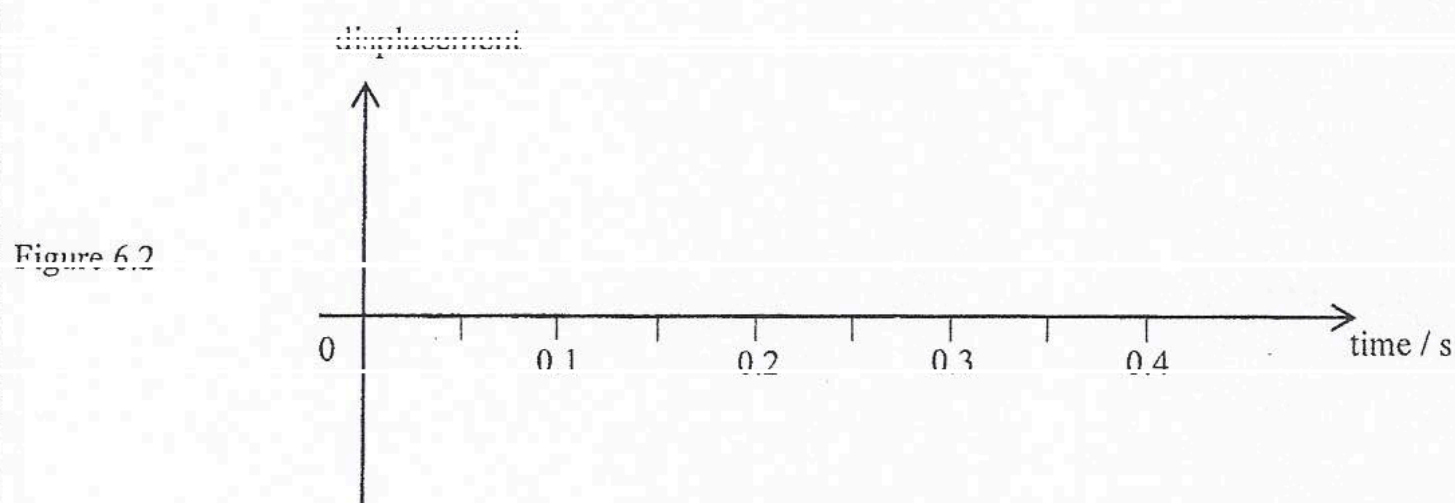
6. (a) A dipper vibrating with a frequency of 5 Hz is put in a water tank. Figure 6.1 shows the displacement-distance graph of the water wave at time $t = 0$. Y is a particle in the water tank.



- (i) Determine the wave speed of the water wave. (2 marks)

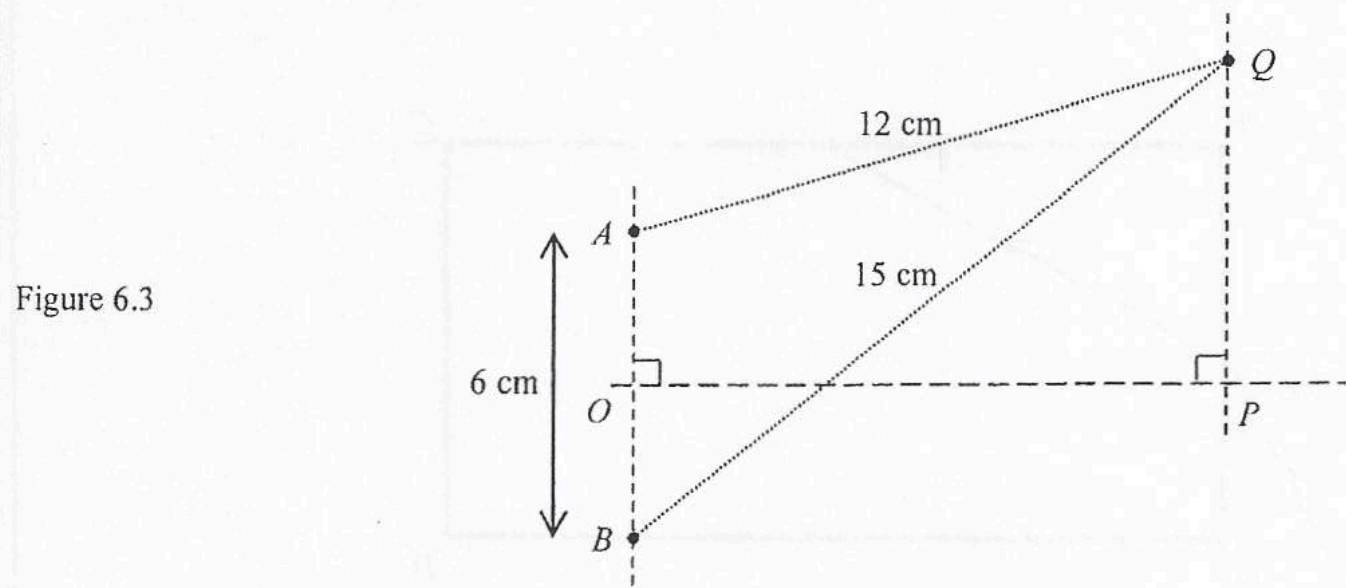
- (ii) State the direction of motion of particle Y at $t = 0$. (1 mark)

- (iii) Sketch the displacement-time graph of particle Y between $t = 0$ and $t = 0.4$ s in Figure 6.2. (2 marks)



Answers written in the margins will not be marked.

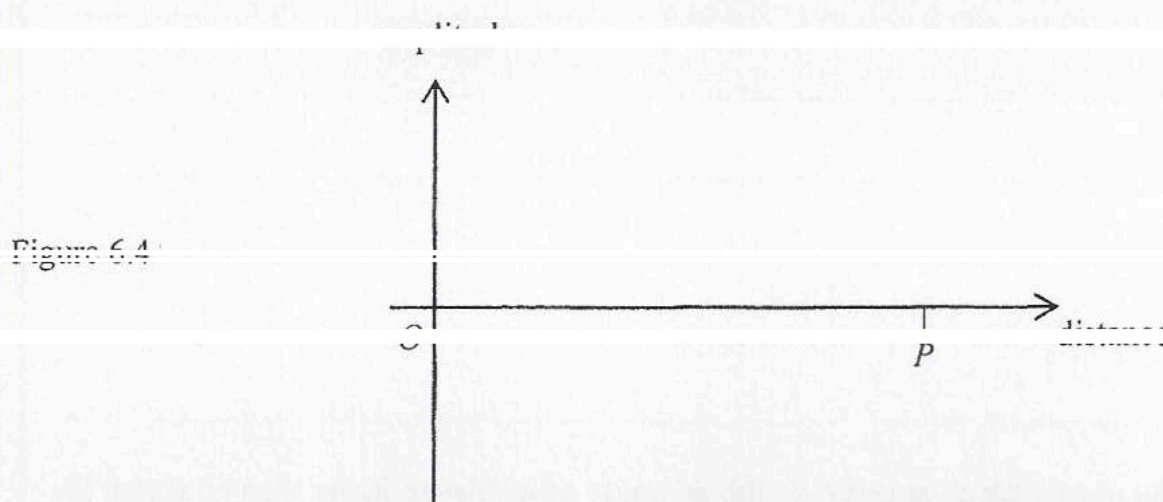
- (b) In Figure 6.3, A and B are two dippers vibrating in phase in a water tank. The distance between A and B is 6 cm. OP is the perpendicular bisector of AB . Q is a second minimum from P , where $AQ = 12$ cm and $BQ = 15$ cm.



- (i) Explain why a minimum occurs at Q . (2 marks)

- (ii) Determine the wavelength of the water wave. (2 marks)

- (iii) Sketch in Figure 6.4 how the **AMPLITUDE** of the water wave varies along the line OP . (1 mark)



Answers written in the margins will not be marked.

5.

Figure 5.1

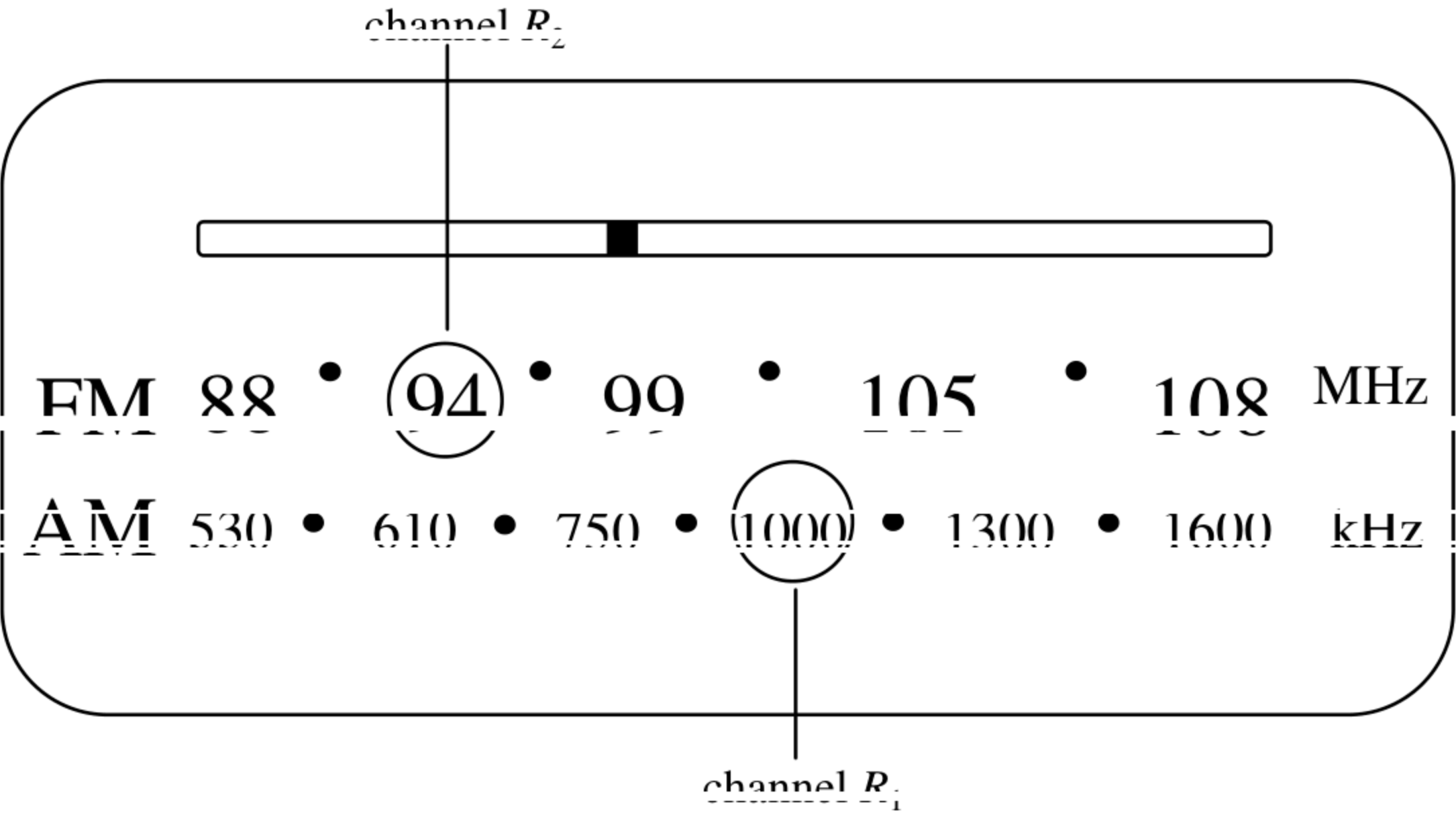


Figure 5.1 shows the display panel of a radio and the broadcasting frequencies of two radio channels R_1 and R_2 . Given : speed of electromagnetic waves = $3.00 \times 10^8 \text{ m s}^{-1}$

- (a) Find the wavelength of the radio waves used by channel R_1 . (1 mark)

.....

.....

.....

.....

.....

- (b) Anita’s house is surrounded by hills and at her house, the reception of one of the two radio channels is better. For which radio channel is the reception better ? Explain your answer. (2 marks)

.....

.....

.....

.....

.....

Answers written in the margins will not be marked.